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SAStainDiff: Self-supervised stain normalization by stain augmentation using denoising diffusion probabilistic models

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ABSTRACT

With the development of computer-aided detection/diagnosis, histopathological images become increasingly important for cancer diagnosis and prognosis. However, different stain styles in histopathological images arise from the difference in stain techniques, operator skills, and scanner specifications. These stain styles reduce the robustness of computer-aided detection/diagnosis algorithms. Existing stain normalization methods often suffer from poor generalization ability and the issue of information loss. In this paper, we propose a new self-supervised diffusion probabilistic modeling approach for stain normalization with stain augmentation training strategy and rescheduled sampling strategy, termed SAStainDiff. Specifically, we employ stain augmentation to simulate different stain styles and learn any stain distribution through diffusion models in a self-supervised manner while preserving the histopathological structure. We employ rescheduled sampling strategy that selects fewer sampling step sizes and a different initial sampling point. This reduces the inference time, which is comparable to mainstream methods, while keeping the performance. We conduct experiments on mutual stain normalization between breast cancer images scanned by two different scanners. Additionally, we explore the application of stain normalization in lymphoma classification and colon gland segmentation. Experimental results demonstrate that our method exhibits excellent generalization capabilities and adapts well to different tissue textures and stain styles without retraining, achieving satisfactory performance in terms of both speed and quality. Our proposed SAStainDiff method can improve the accuracy of disease diagnosis and subsequent analysis, ultimately benefiting clinical practice and advancing medical research. The code and sample data are publicly available on GitHub.

1. Introduction

Digital pathology images involve capturing entire glass slides using scanners and storing them as digital images. Most tissues and cells under the microscope appear transparent, necessitating the usage of stains to process tissue samples [1]. However, different stain styles typically exist in pathology images processed by various laboratories and institutions. Factors such as different stain agents, stain protocols, operational procedures, and scanner specifications contribute to color and intensity styles in histopathological images, affecting the final appearance of the digital pathology images. The different stain styles in histopathological images give rise to several challenges. Without pre-processing the images, diagnostic results may be incorrect [2]. In recent years, computer-aided diagnosis (CAD) has emerged as one of the most active research areas in digital pathology image analysis. However, the performance of CAD systems is significantly hindered by variations in color and intensity styles, which impedes the application of CAD in histopathology [3]. Therefore, stain normalization is usually an integral component in designing a stable and reliable CAD system [4].

Till now, numerous methods have been developed for stain normalization of histopathological images [5,6], including traditional mathematical and learning-based methods. As for traditional approaches, Reinhard [7], Macenko [8], Vahadane [9], and others [10–13] have employed mathematical techniques such as linear transformations, singular value decomposition of the optical density, and sparse non-negative matrix factorization, to calculate stain vectors. Currently, most learning-based stain normalization methods are based on GANs, which

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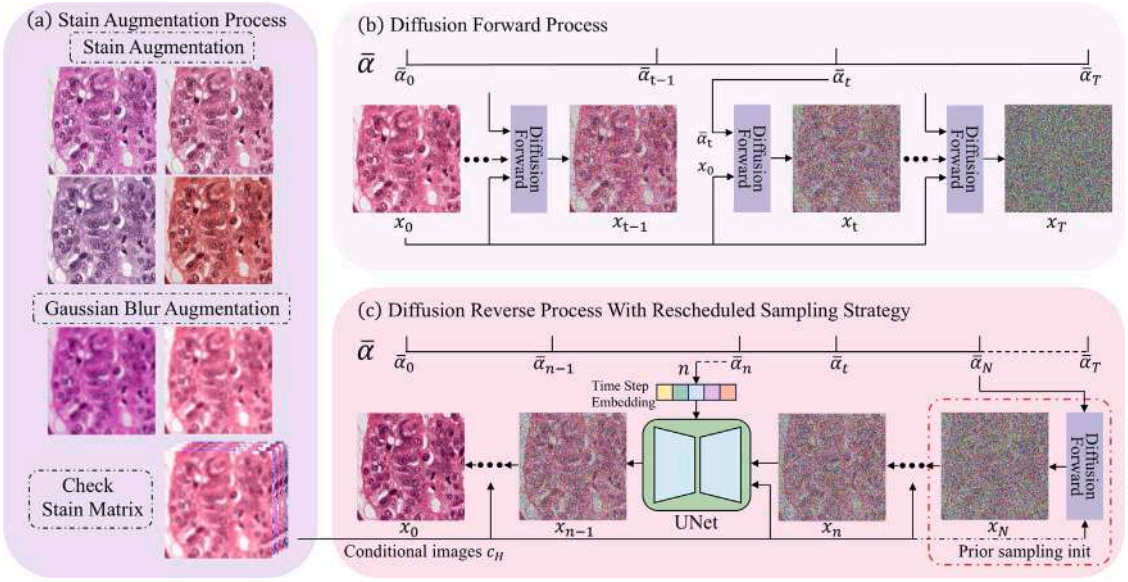


Fig. 1. Framework of the proposed SAStainDiff. (a) The stain augmentation scheme enhances the image by randomly perturbing color, adding Gaussian blur, and then choosing the augmented results that meet the requirements based on data statistic characteristics. (b) The diffusion forward process involves gradually adding continuous noise to clear image x_0. (c) The diffusion reverse process involves incorporating the conditional images c_H with the noisy image x_n into the UNet, to remove the noise gradually. When using the rescheduled sampling strategy, a new initial sampling point N is chosen to incorporate the prior information into the sampling process and reduce the number of steps in subsequent sampling.

can be broadly categorized into supervised, unsupervised, and self-supervised learning methods. Supervised methods rely on paired images [14]. Unsupervised methods primarily rely on CycleGAN [15] and its variants [16–18], while Tosta considers both handcrafted and learning features with dictionary learning [19]. Self-supervised stain normalization methods focus on recoloring grayscale images back to original RGB space [20–22]. Despite the advancements achieved, limitations still persist, where information loss remains a prominent issue.

In practice, self-supervision is an effective training strategy in the absence of paired data. However, most approaches employing self-supervised frameworks for stain normalization merely separate them into different color vectors and convert color images into grayscale images [20–23]. These colorization-based methods often risk losing crucial information present in the original histopathological images. Alternatively, stain augmentation allows for perturbation of the color style of histopathological images while preserving the original color information and is also employed as a means to improve model generalization [24]. Therefore, employing a stain augmentation-based self-supervised training method to learn the normalization capability for different stain styles is a more effective approach.

Compared with GANs, Denoising Diffusion Probabilistic Models (DDPMs) have also demonstrated strong performance in various image-related tasks in recent years [25,26]. However, only a few works have applied DDPMs to stain normalization. Jeong [23] utilizes a score-based diffusion model to separate histological images into Hematoxylin and Eosin (H&E) channels, which were then individually normalized and recombined. Shen [27] designs a cycle-consistent diffusion model capable of learning from unpaired histological images. These methods require substantial computational resources and suffer from slow inference speeds.

Therefore, in contrast to existing methods, we innovatively design a self-supervised color-to-color stain normalization algorithm based on diffusion models with a stain augmentation training strategy and rescheduled sampling strategy. The main contributions are twofold:

- We systematically transform the stain normalization task into a self-supervised formulation of the pixel-wise color-to-color image mapping task. This stain augmentation training strategy is flexible, practical, and does not require retraining when applied to different datasets.
- We propose a new self-supervised diffusion probabilistic modeling approach for stain normalization with a rescheduled sampling strategy that improves the sampling speed while keeping the performance.

2. Methods

In this section, we first introduce the stain augmentation strategy. Subsequently, we present the conditional diffusion model. Finally, we elaborate on the design of our rescheduled diffusion model. The proposed rescheduled conditional diffusion model framework incorporates a self-supervised stain augmentation approach for stain normalization called SAStainDiff, as shown in Fig. 1.

2.1. Histopathological images stain augmentation

The training paradigm of conditional diffusion model relies on paired images, which does not align with the practical scenario of histopathological images. To overcome the challenge of acquiring pixel-aligned data in actual clinical practice and achieving normalization for arbitrary stain styles, we employ a self-supervised learning approach.

During the training process, we perform stain augmentation [28,29] on the original image x_0 to simulate paired histopathological images with different stain styles, as depicted in Fig. 1(a). Consequently, in the inference phase, we can utilize any stain-style image as a condition to standardize it to the target stain distribution. The stain augmentation process can be summarized as Algorithm 1, where the $matrix_i, i \in \{1, 2, \ldots, n\}$ refers to the chromaticity matrix extracted from histopathological images using the Macenko algorithm [8]. The Stain Database consists of images that require stain normalization. Parameters σ_1 and σ_2 are empirical values that determine the degree of stain augmentation. In this study, we set both parameters to 0.7.

The stain augmentation process consists of two steps. In the first step, we randomly select the Hematoxylin and Eosin stain matrices in the Stain Database and perturb them to simulate various stain styles. The outliers are filtered out to ensure the generated stain styles lie in the distribution of the Stain Database. Specifically, we search for the k-nearest neighbors of the augmented matrices $matrix_{c_H}$ within an Euclidean distance of 0.1 in the stain database to determine their

Algorithm 1: Stain Augmentation

Input: Target images x_0, Parameters σ_1 and σ_2 ;
$Stain\ Database\left[matrix_1, matrix_2, \cdots, matrix_n\right]$
Output: Augmented images c_H

do
  $a \sim uniform\left(1-\sigma_1, 1+\sigma_1\right)$;
  $b \sim uniform\left(-\sigma_2, \sigma_2\right)$;
  $matrix_{c_H} \sim uniform\left(Stain\ Database\right)$;
  $matrix_{c_H} \leftarrow a \times matrix_{c_H} + b$;
while $matrix_{c_H}$ *is an outlier*;
$c_H \leftarrow$ Macenko normalize$\left(x_0, matrix_{c_H}\right)$;
$c_H \leftarrow Gaussian\ blur\left(c_H\right)$;

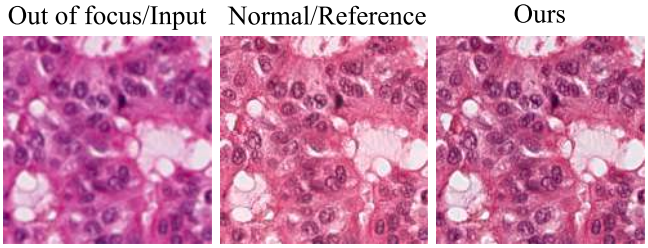


Fig. 2. Effect of Gaussian blur augmentation. The image "Out of focus/Input" demonstrates the presence of image blur, while the image "Ours" showcases the results of our stain normalization with Gaussian blur augmentation, enhancing the image clarity.

acceptability. It is accepted if the minimum number of neighbors exceeds 5; otherwise, the augmentation process is repeated. In the second step, we apply Gaussian blur to the augmented images. As shown in Fig. 2, scanners occasionally suffer from inaccurate focusing, leading to blurred images. So, we aim to incorporate blur degradation during the augmentation process and leverage the powerful restoration capability of the diffusion model to improve image quality. Through this approach, we obtain a well-defined stain augmentative image, allowing us to adjust the model accordingly based on specific stain distributions within the dataset, as needed.

2.2. DDPM with guided histopathological images

This section aims to design a diffusion model for the stain normalization task. Denoising Diffusion Probabilistic Model (DDPM) [30] is mostly used for image generation and exhibits high randomness. Therefore, we utilize an image-to-image conditional diffusion model as described in the work of Palette [31].

The conditional diffusion model consists of two processes: the forward process and the reverse process. The forward process aims to perturb the data until its distribution approaches a standard Gaussian distribution. The reverse process attempts to recover the original data distribution from a given prior distribution. Both of these processes are modeled as Markov chains.

In the forward process $q\left(x_t \mid x_0\right)$, the image x_t can be represented by adding noise ϵ to the original image x_0:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \tag{1}$$

In the reverse process $p_\theta\left(x_{t-1} \mid x_t, c_H\right)$, the previous image x_{t-1} can be predicted by the current image x_t, conditional image c_H and time step t. To generate the normalization image, the conditional diffusion model concatenates x_t with c_H as input to the UNet model, which is responsible for noise $\hat{\epsilon}_\theta\left(x_t, c_H, t\right)$ prediction. The conditional reverse process can be represented as follows:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\epsilon}_\theta\left(x_t, c_H, t\right)\right) + \sigma_t z \tag{2}$$

Algorithm 2: Diffusion Training

Input: target images x_0 $(Target\ Domain)$
Output: Parameters of the UNet $\hat{\epsilon}_\theta$

repeat
  $c_H \leftarrow Stain\ Augmentation\left(x_0\right)$;
  $t \sim Uniform\left(\{1, ..., T\}\right)$;
  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$;
  $x_t \leftarrow \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$;
  Take gradient descent step on: $\nabla_\theta \| \epsilon - \hat{\epsilon}_\theta\left(x_t, c_H, t\right) \|^2$
until *converged*;

where $z \sim \mathcal{N}(0, \mathbf{I})$ is only used when $t > 1$, α_t, $\bar{\alpha}_t$ and σ_t are parameters for diffusion model.

In the training process, the L2 loss function is employed to fit the noise $\hat{\epsilon}_\theta\left(x_t, c_H, t\right)$ predicted by the UNet model and the noise ϵ added from observation image x_0 to noise image x_t:

$$Loss = \| \epsilon - \hat{\epsilon}_\theta\left(x_t, c_H, t\right) \|^2 \tag{3}$$

The diffusion training process with stain augmentation strategy can be summarized as Algorithm 2.

Moreover, in the inference process, histopathological images are typically of flexible and large size, while most methods can only generate fixed-size images. The common approach is overlapping sliding window patches. However, due to styles in stain intensity and tissue structure across different patches, there can be significant color discrepancies. This only mitigates the issue of patch color inconsistency to some extent. We reference the approach Restore Any-size Image [32] and make adaptations specifically for histopathological images. Specifically, in the inference process of the diffusion model, which involves T steps of sampling, we perform overlapping sliding window patches for each sampling step, as shown in Fig. 3. Compared with the post-fusion strategy, fusion during inference can eliminate color inconsistency artifacts better.

2.3. Rescheduled sampling strategy

Denoising Diffusion Probabilistic Model (DDPM) [30] suffers from long inference times. In this section, we design a rescheduled sampling strategy for the stain normalization task that can reduce inference time without compromising image quality.

Denoising Diffusion Implicit Model (DDIM) [33] is an effective strategy for reducing sampling steps. Unlike Eq. (2), the DDIM reverse process is a non-Markovian diffusion process and can be represented as follows, where x_{prev} and x_t are not required to be adjacent.

$$\begin{aligned} x_{prev} = & \sqrt{\bar{\alpha}_{prev}}\left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\hat{\epsilon}_\theta\left(x_t, c_H, t\right)}{\sqrt{\bar{\alpha}_t}}\right) \\ & + \sqrt{1-\bar{\alpha}_{prev}}\hat{\epsilon}_\theta\left(x_t, c_H, t\right) \end{aligned} \tag{4}$$

However, relying solely on DDIM performs poorly in stain normalization tasks. When there are too few sampling steps, the image colorization is inaccurate, and the quantitative evaluation of the images significantly deteriorates. However, the sampling time is too long when there are too many sampling steps. To address this conflict, following our previous work [34], we propose a rescheduled sampling strategy that enables a diversified noise schedule and sampling steps, improving image quality.

Specifically, outputs of the original sampling strategy are sampled from pure Gaussian noise. However, for stain normalization tasks, the structures and color information are totally lost if we start sampling from pure Gaussian noise. Therefore, we advocate selecting a suitable initial sampling point x_N where the noised input image and the noised target image have the same distribution. Subsequently, N values of $\bar{\alpha}_n$

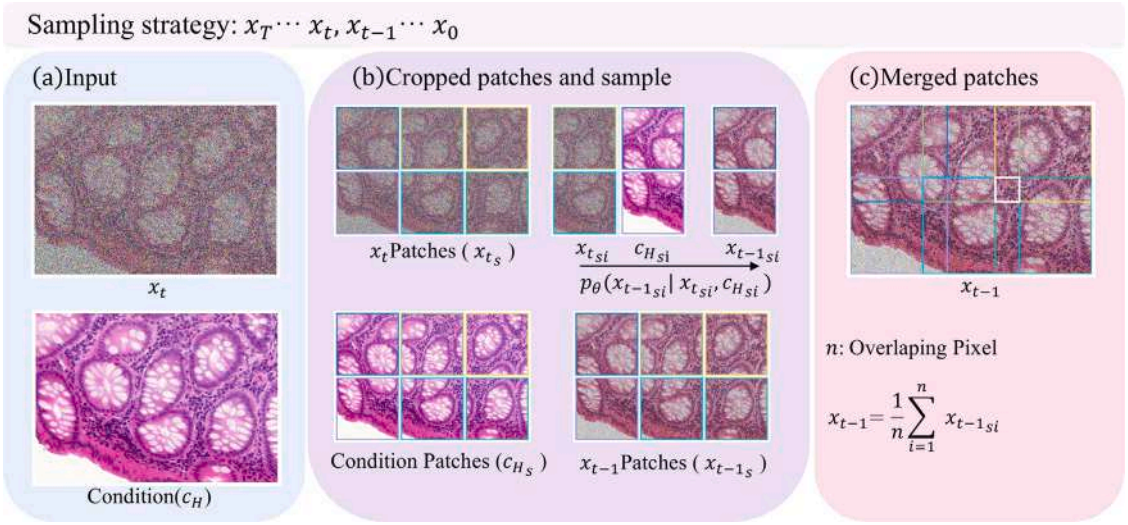


Fig. 3. The sampling strategy to address the color variations is illustrated as follows: (a) Denotes the capability to input high-resolution images. (b) For each times step t, the image is partitioned into fixed-size patches using overlapping sliding windows, and the corresponding image at the next time step is predicted on these patches. (c) Merged patches and the overlapping pixels are averaged.

Algorithm 3: Rescheduled Sampling

Input: conditional images c_H (*Source Domain*), step size N, start noise level $\bar{\alpha}_N$, rescheduled noise schedule $\bar{\alpha}_{\{1:N\}}$
Output: Target stain normalization image x_0

$\epsilon \sim \mathcal{N}(0, \mathbf{I})$;
$x_N \leftarrow \sqrt{\bar{\alpha}_N} c_H + \sqrt{1-\bar{\alpha}_N}\epsilon$;
for $n = N, ..., 1$ **do**
  $z \sim \mathcal{N}(0, \mathbf{I})$ if $t > 1$, else $z \leftarrow 0$;
  $x_{n_s}, c_{H_s} \leftarrow overlap\ slid\ window\left(x_n, c_H\right)$;
  $x_{n-1_s} \leftarrow \sqrt{\bar{\alpha}_{n-1}}\left(\frac{x_{n_s} - \sqrt{1-\bar{\alpha}_n}\hat{\epsilon}_\theta\left(x_n, c_{H_s}, \bar{\alpha}_n\right)}{\sqrt{\bar{\alpha}_n}}\right) + \sqrt{1-\bar{\alpha}_{n-1}}\hat{\epsilon}_\theta\left(x_n, c_{H_s}, n\right)$;
  $x_{n-1} \leftarrow merge\left(x_{n-1_s}\right)$;
end

are chosen from this noise level $\bar{\alpha}_N$ to $\bar{\alpha}_1$, with N representing the sampling step size. Iterations are then performed from $\bar{\alpha}_N$ onwards, and DDIM is employed for subsequent sampling within the selected values of $\bar{\alpha}_n$. By incorporating the information of the conditional images into the diffusion model through this prior, we can improve the generative performance of the model while reducing the sampling steps. Furthermore, we observe that different step sizes and initial sampling points also affect the intensity of stain normalization, allowing for selecting suitable step sizes and initial sampling points based on specific circumstances to achieve a desired level of stain normalization similarity with the reference image. The proposed rescheduled sampling strategy can be summarized as Algorithm 3. The rescheduled noise schedule $\bar{\alpha}$ represents the set of values $\bar{\alpha}_n$ we have chosen. In this work, the optimal rescheduled noise schedule $\bar{\alpha}$ may vary for different tasks and models. We set a universal $\bar{\alpha}$ started from $\bar{\alpha}_{900}$ in original 1000 time steps and 5 step size at equal intervals.

3. Experiment settings

We evaluate SAStainDiff on three public datasets. Firstly, the MITOS-ATYPIA'14 Challenge dataset[1] is used to assess the performance of stain normalization both quantitatively and qualitatively. Secondly, the Camelyon16 dataset [35] is employed to evaluate the effect of stain normalization in the classification task. Lastly, the MICCAI'15 GlaS challenge dataset [36] is utilized to evaluate the effect of stain normalization in the segmentation task. These three datasets encompass tissue sections from two prevalent diseases, breast cancer and colorectal cancer, scanned by five different scanners, thereby effectively capturing the diversity of stain styles encountered in actual clinical practice.

To ensure fairness, for both the classification and segmentation tasks, we conduct training and testing on the stain normalization results of different methods, using the same dataset partition, network settings, and target domain. All experiments are conducted using Python 3.8.17 and torch 1.13.1 on a single NVIDIA A100 80 GB GPU.

3.1. Stain normalization

Dataset. The MITOS-ATYPIA'14 dataset consists of the same breast cancer biopsy slides scanned by two scanners: Aperio Scanscope XT and Hamamatsu Nanozoomer 2.0-HT. We perform mutual conversion of domains between these two scanners, normalizing the images from the Hamamatsu domain to the Aperio domain and normalizing the images from the Aperio domain to the Hamamatsu domain. Officially, there are 424 images available at a magnification of 20×. For the Aperio domain, the image size is 1539 × 1376, and the resolution is 0.4910 μm/pixel. For the Hamamassu domain, the image size is 1663 × 1485, and the resolution is 0.454598 μm/pixel in horizontal and 0.455062 μm/pixel in vertical. There is a slight misalignment between these two domains. Therefore, we register Hamamatsu to Aperio using SIFT [37] and the RANSAC algorithm. In the training process, we randomly crop patches sized of 256 × 256 from 284 of the 424 images, while in the testing process, we crop 3500 patches sized of 256 × 256 without overlap from the rest 140 images as test sets. The percentage split between the training and test sets is 67% and 33%. We do not partition a validation set in this experiment. It is important to note that our self-supervised approach solely utilizes images from the target domain during training, with the paired data solely for evaluating the performance of our proposed method.

Implementation details. We employ the Adam optimizer with a learning rate of 1×10^{-4}, a batch size of 16, a diffusion time T of 1000, in total 700,000 training iterations, an exponential moving average of 0.9999, a linear noise schedule and stain augmentation training process Algorithm 2. These parameters, along with other relevant parameters, are empirically set based on previous works [25,31].

[1] https://mitos-atypia-14.grand-challenge.org.

3.2. Stain normalization for classification

Dataset. The Camelyon16 dataset [35] comprises sentinel lymph node whole-slide images scanned by 3DHistech Pannoramic Flash II 250 scanner at the Radboud University Medical Center (RUMC) and Hamamatsu NanoZoomer-XR C12000-01 scanner at the University Medical Center Utrecht (UMCU). The dataset includes both tumor and normal regions. Following the setup in [38], we extract 40,000 patches sized of 256 × 256 from tumor and normal regions of the 170 RUMC's WSIs as well as 10,000 patches from the 50 UMCU's WSIs. The images collected from the two centers exhibit obviously different styles, and we standardize them to the Aperio domain.

Since we want to test the effect of stain normalization in the classification task, it is better to conduct cross-domain experiments, i.e., training the classifier on the RUMC domain and testing it on the UMCU domain. The images collected from RUMC are used as the training and validation set for the classifier, with 38,000 images in the training set and 2000 images in the validation set. The images collected from UMCU are entirely used as the test set to evaluate the performance of the standardized classifier. The percentage split between the train/validation/test sets is 76%, 4%, and 20%.

Implementation details. We employ the SqueezeNet V1.1 model, Adam optimizer, cross-entropy loss function, a learning rate of 2×10^{-4}, a batch size of 64, and a total of 60 training epochs for all stain normalization methods. Based on the validation set loss, we adjust the learning rate using ReduceLROnPlateau with patience of 10 and a factor of 0.1. These parameters are set empirically according to [38].

3.3. Stain normalization for segmentation

Dataset. The MICCAI'15 GlaS dataset [36] is a collection of images used for the segmentation task of colorectal adenocarcinoma. It consists of 165 images scanned by Zeiss MIRAX MIDI Slide Scanner. We resize all the images to 512 × 512 and conduct a 5-fold cross-validation. In each fold, the percentage split for train/validation/test is 64%, 16%, and 20%.

Implementation details. We employ the UNet model, Adam optimizer, dice loss function, a learning rate of 1×10^{-3}, a batch size of 2, and a total of 200 training epochs for all stain normalization methods. Based on the validation set loss, we adjust the learning rate using ReduceLROnPlateau with patience of 20 and a factor of 0.1. These parameters are set empirically according to [39].

3.4. Evaluation metrics

The metrics used for quantitative comparison of stain normalization are several widely-used full-reference metrics, including Structural Similarity Index Measure (SSIM) [40], Feature Similarity Index for Image Quality Assessment (FSIM) [41], Multi-Scale Structural Similarity Index Measure (MS-SSIM) [42], Peak Signal-to-Noise Ratio (PSNR) and Learned Perceptual Image Patch Similarity (LPIPS) [43]. LPIPS employs AlexNet for computation.

Moreover, for the classification task, we evaluate the performance using metrics including Accuracy, F1 score, and the area under the ROC curve (AUC). For the segmentation task, the evaluation is based on metrics including Pixel Accuracy, Dice score, and Hausdorff Distance (HD).

3.5. Competitive methods

Since some of the previous works do not provide their source code, it is inconvenient to perform direct comparisons. We only select 11 representative methods for comparison, including Reinhard [7], Macenko [8], Vahadane [9], StainGAN [15], SIST [20], SAGAN [44], CAGAN [22], MultiPathGAN [16], JeongSDE [23], Pix2Pix [45] and Palette [31]. JeongSDE does not have a publicly available code, and we re-implement it and conduct experiments based on its original training settings. Due to the extensive computational time required for JeongSDE and Palette inference, we do not validate JeongSDE in the classification task, and Palette is validated using a DDIM sampler with 25 steps in the classification task. Pix2Pix and Palette are image-to-image translation models that require paired images, so we train them with our stain augmentation training strategy to evaluate the applicability of our formulation. All parameters are set according to their respective papers, and we train them using the same dataset as the proposed method.

4. Results and discussion

4.1. Stain normalization on MITOS-ATYPIA'14

The quantitative results for stain normalization on the MITOS-ATYPIA'14 dataset are presented in Tables 1 and 2. As can be seen, Pix2Pix, Palette, and Our model, trained with stain augmentation training strategy, outperform most existing methods, demonstrating superior competitive performance. This substantiates the effectiveness of our self-supervised stain augmentation training approach for stain normalization.

The computational resources of the competitive methods are presented in Table 3. The inference time refers to the average time taken for performing inference on 10 images of size 256 × 256, with 5 repetitions. From the result, due to the nature of network design, GANs exhibit significantly faster training and inference speeds, while DDPMs require a larger computational resource investment. However, under our proposed rescheduled sampling strategy, we achieve a 120-times increase in inference speed than the original Palette model, with virtually no degradation in image quality. Additionally, a significant improvement is observed in the experiment of normalizing slices from Aperio to Hamamatsu. This is because the fully sampled normalized images exhibit deeper staining in this experiment, and through rescheduled resampling adjustments, the number of sampling steps can be reduced while controlling the intensity of staining, resulting in a closer resemblance to the reference images. This demonstrates that our proposed SAStainDiff is an effective approach in the context of stain normalization tasks, offering promising performance and speed.

Qualitative evaluation is shown in Figs. 4 and 5, where the difference maps of all methods are normalized to [0,1] together for better visualization. As depicted in Fig. 4, apart from StainGAN, MultiPathGAN, and the model trained with our proposed stain augmentation training strategy, the results of other methods alter the original color of red blood cells. Among them, StainGAN and MultiPathGAN exhibit lighter colors and decreased contrast. In comparison, our proposed stain augmentation training strategy yields the best visual results. Furthermore, in Fig. 5, the results of Reinhard, JeongSDE, and Palette exhibit staining errors, displaying excessively dark or anomalous color bands. The images normalized by StainGAN, SIST, SAGAN, CAGAN, and MultiPathGAN exhibit lighter coloration, indicating incomplete stain normalization. When compared with the reference images, noticeable color deviations can be observed. An anomaly is also observed at the top left corner of the SAGAN. In contrast, our method achieves a closer color and texture resemblance to the reference image than the competitive methods. This observation highlights the strong tissue recognition capability of our model. Our proposed SAStainDiff method achieves the ability to normalize stain to the target distribution while better preserving and enhancing the original texture information.

Overall, the stain augmentation training strategy is a highly effective self-supervised training method for stain normalization. Our proposed SAStainDiff balances speed and quality, resulting in the highest ratings.

Table 1
Performance of stain normalization on MITOS-ATYPIA'14, from Hamamatsu to Aperio. The best result is in **bold** and the second best result is underlined. The methods ranking highly in various metrics are predominantly concentrated on those trained with our stain augmentation strategy (Pix2Pix, Palette, and Ours).

Hamamatsu → Aperio	SSIM (↑)	MS-SSIM (↑)	FSIM (↑)	PSNR (↑)	LPIPS (↓)
Reinhard [7]	0.742 ± 0.094	0.928 ± 0.039	0.851 ± 0.040	18.435 ± 2.507	0.177 ± 0.042
Macenko [8]	0.722 ± 0.129	0.920 ± 0.051	0.850 ± 0.046	20.326 ± 2.771	0.154 ± 0.055
Vahadane [9]	0.729 ± 0.122	0.920 ± 0.057	0.850 ± 0.045	20.575 ± 2.557	0.159 ± 0.054
StainGAN [15]	0.736 ± 0.100	0.930 ± 0.036	0.860 ± 0.036	**22.044 ± 1.521**	0.125 ± 0.040
SIST [20]	0.760 ± 0.097	0.941 ± 0.034	0.863 ± 0.040	21.720 ± 1.298	0.144 ± 0.042
SAGAN [44]	0.742 ± 0.095	0.928 ± 0.035	0.853 ± 0.036	21.108 ± 1.259	0.170 ± 0.034
CAGAN [22]	0.742 ± 0.095	0.934 ± 0.036	0.860 ± 0.038	21.037 ± 1.237	0.153 ± 0.042
MultiPathGAN [16]	0.751 ± 0.099	0.933 ± 0.036	0.861 ± 0.038	21.533 ± 1.438	0.136 ± 0.042
JeongSDE [23]	0.756 ± 0.093	0.940 ± 0.033	0.861 ± 0.039	20.870 ± 1.307	0.158 ± 0.049
Pix2Pix [45]	0.754 ± 0.102	0.940 ± 0.036	**0.866 ± 0.038**	21.910 ± 1.462	**0.122 ± 0.050**
Palette [31]	**0.761 ± 0.099**	**0.942 ± 0.034**	0.866 ± 0.039	21.673 ± 1.644	0.132 ± 0.046
SAStainDiff(Ours)	0.759 ± 0.100	0.941 ± 0.034	0.865 ± 0.039	21.861 ± 1.483	0.130 ± 0.046

Table 2
Performance of stain normalization on MITOS-ATYPIA'14, from Aperio to Hamamatsu. The best result is in **bold** and the second best result is underlined. Apart from LPIPS, SAStainDiff(Ours) achieved the best performance across all metrics. Pix2Pix and Palette also achieved favorable results compared to other methods except for StainGAN.

Aperio → Hamamatsu	SSIM (↑)	MS-SSIM (↑)	FSIM (↑)	PSNR (↑)	LPIPS (↓)
Reinhard [7]	0.735 ± 0.102	0.930 ± 0.039	0.850 ± 0.044	18.075 ± 2.516	0.173 ± 0.040
Macenko [8]	0.708 ± 0.109	0.922 ± 0.042	0.830 ± 0.050	20.742 ± 2.044	0.167 ± 0.055
Vahadane [9]	0.732 ± 0.107	0.926 ± 0.040	0.843 ± 0.049	21.079 ± 2.160	0.157 ± 0.052
StainGAN [15]	0.787 ± 0.090	0.944 ± 0.033	0.872 ± 0.037	23.059 ± 1.709	**0.122 ± 0.039**
SIST [20]	0.773 ± 0.093	0.938 ± 0.032	0.867 ± 0.037	21.169 ± 1.142	0.154 ± 0.037
SAGAN [44]	0.765 ± 0.093	0.928 ± 0.035	0.859 ± 0.039	22.245 ± 1.843	0.140 ± 0.046
CAGAN [22]	0.748 ± 0.098	0.923 ± 0.038	0.844 ± 0.042	21.401 ± 1.414	0.184 ± 0.052
MultiPathGAN [16]	0.765 ± 0.098	0.938 ± 0.036	0.861 ± 0.043	22.716 ± 1.752	0.138 ± 0.044
JeongSDE [23]	0.753 ± 0.098	0.937 ± 0.035	0.853 ± 0.042	21.529 ± 1.765	0.150 ± 0.046
Pix2Pix [45]	0.764 ± 0.097	0.939 ± 0.034	0.859 ± 0.042	22.403 ± 1.695	0.136 ± 0.046
Palette [31]	0.781 ± 0.094	0.941 ± 0.033	0.869 ± 0.040	21.909 ± 2.148	0.131 ± 0.045
SAStainDiff(Ours)	**0.791 ± 0.092**	**0.947 ± 0.032**	**0.875 ± 0.039**	**23.523 ± 1.808**	0.123 ± 0.043

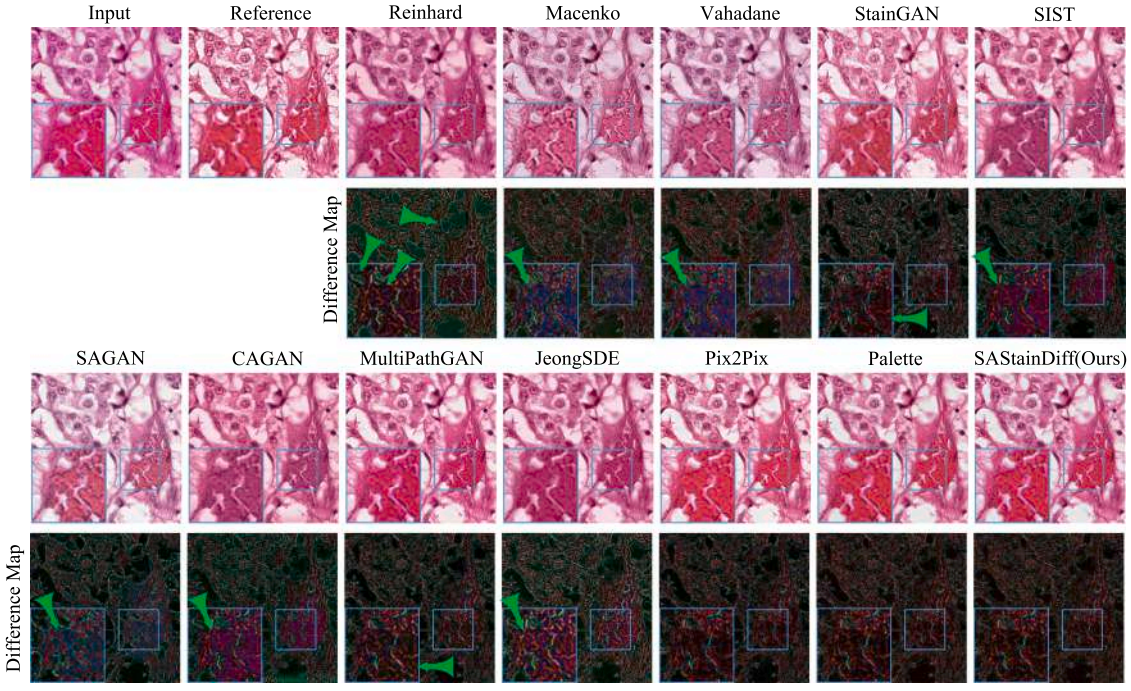


Fig. 4. Visual comparison of stain normalization results on the MITOS-ATYPIA'14 dataset, from the Hamamatsu domain to the Aperio domain. From the difference map indicated by the arrow, it is evident that some methods result in the loss of certain color information or incorrectly generated colors, whereas our method demonstrates superior fidelity and ability.

4.2. Stain normalization in the classification task

In the Camelyon16 [35] classification task, to validate the generalization of our proposed method on other datasets, we perform stain normalization of images from the RUMC and UMCU domains to the Aperio domain of MITOS-ATYPIA'14. Except for methods trained with our proposed stain augmentation training strategy and the grayscale stain methods, all other methods require retraining. Since we want

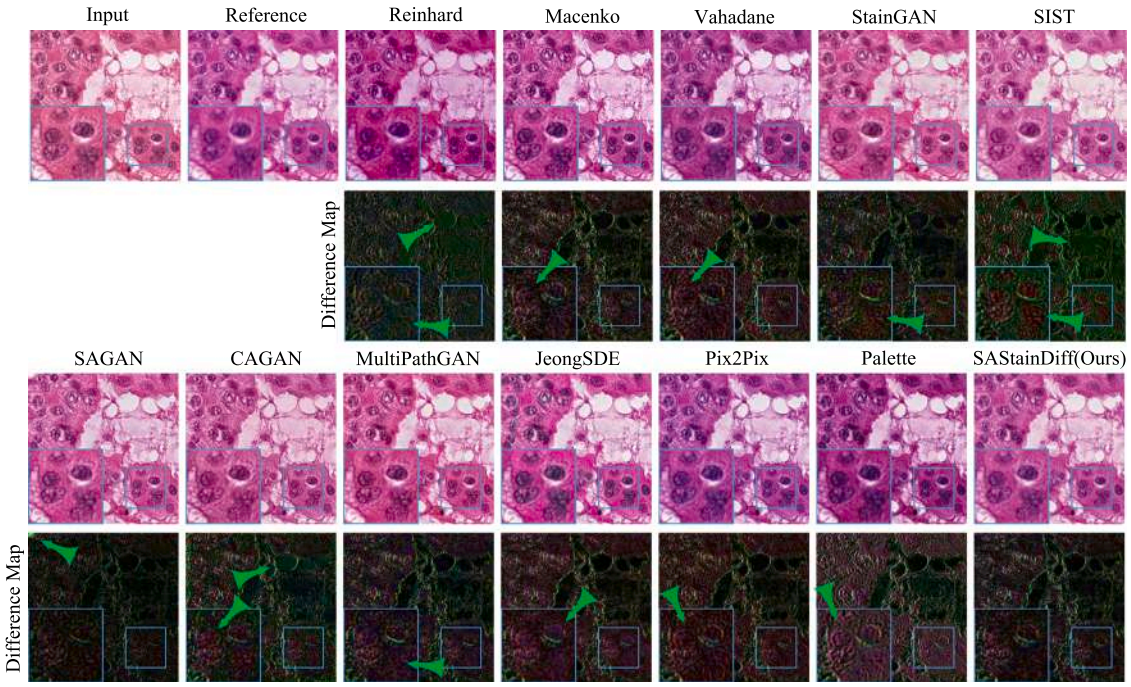


Fig. 5. Visual comparison of stain normalization results on the MITOS-ATYPIA'14 dataset, from the Aperio domain to the Hamamatsu domain. From the difference map indicated by the arrow, some methods result in images with noticeable color deviations. In contrast, our proposed method with the rescheduled sampling strategy (specifically Palette and Ours), exhibits a closer resemblance to the reference.

Table 3
Training and inference time. GANs have advantages in both training and inference. DDPMs require substantial resources, but our method has optimized the inference speed, achieving results comparable to Macenko.

Method	Training time (day)	Inference time (s)
Reinhard [7]	–	0.075
Macenko [8]	–	3.555
Vahadane [9]	–	13.745
StainGAN [15]	0.172	1.040
SIST [20]	0.014	3.048
SAGAN [44]	0.022	1.734
CAGAN [22]	0.038	1.667
MultiPathGAN [16]	2.021	1.672
JeongSDE [23]	5.930	1722.605
Pix2Pix [45]	0.164	0.997
Palette [31]	11.677	476.225
SAStainDiff(Ours)	11.677	3.988

Table 4
Comparison of stain normalization performance on Camelyon16 classification task, where "Original" denotes the results without any normalization. In the cross-domain experiment (training on RUMC and testing on UMCU), stain normalization significantly improves results, with Palette and Our method exhibiting the most notable improvement.

Method	Accuracy (↑)	F1 score (↑)	AUC (↑)
Original	0.476	0.058	0.734
Reinhard [7]	0.778	0.774	0.863
Macenko [8]	0.794	0.790	0.878
Vahadane [9]	0.778	0.738	0.875
StainGAN [15]	0.731	0.690	0.828
SIST [20]	0.750	0.714	0.863
SAGAN [44]	0.748	0.709	0.835
CAGAN [22]	0.730	0.662	0.849
MultiPathGAN [16]	0.748	0.737	0.823
Pix2Pix [45]	0.785	0.785	0.859
Palette [31]	**0.806**	0.794	0.888
SAStainDiff(Ours)	0.796	**0.811**	**0.892**

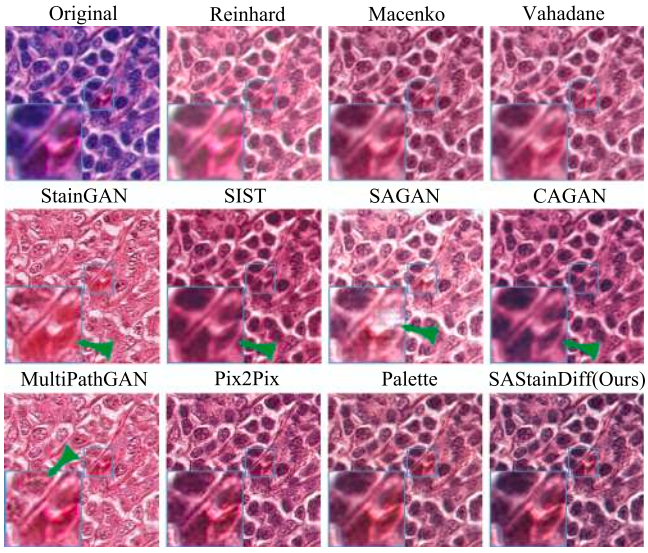


Fig. 6. Visual comparison of stain normalization results on the Camelyon16 dataset. As indicated by the arrow, some methods may produce erroneous results. Our method can produce images with clear structures and pleasing colors.

to test the effect of stain normalization in the classification task, it is better to conduct cross-domain experiments, i.e., training the classifier on the RUMC domain and testing it on the UMCU domain. The results of the intra-domain cross-validation experiment are provided in the supplementary material.

Table 4 presents the quantitative results. It can be observed that Palette and our method outperform other methods. Pix2Pix also demonstrates good performance, but compared to Palette, it falls slightly behind, indicating the stability of diffusion models. As shown in Fig. 6, due to the significant differences in cell structures among Camelyon16 and MITOS-ATYPIA'14 datasets, traditional methods may exhibit stain

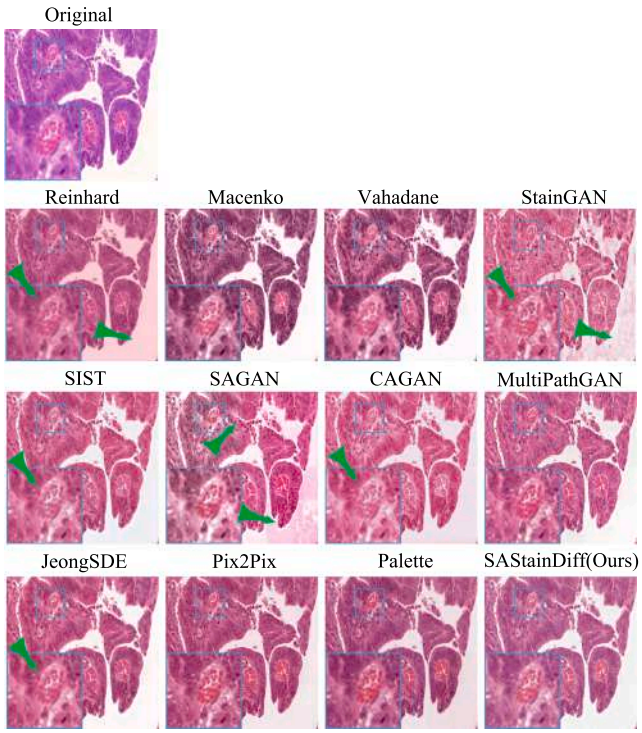


Fig. 7. Visual comparison of stain normalization results on the MICCAI'15 GlaS dataset. As indicated by the arrow, some methods may produce erroneous results. Our method avoids the occurrence of grid artifacts and effectively preserves texture information.

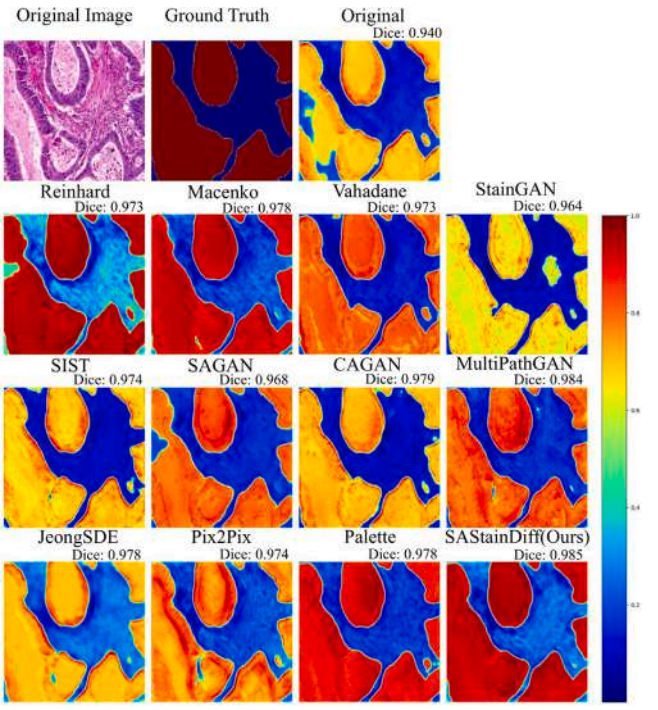


Fig. 8. Visual comparison of softmax in the GlaS segmentation results, where "Original" represents the results without any normalization. The dice scores are computed using a threshold of 0.5. After applying our stain normalization method, superior segmentation performance is demonstrated.

Table 5
Comparison of stain normalization performance in 5-fold cross-validation for the MICCAI'15 GlaS segmentation task, where "Original" represents the results without any normalization. SAStainDiff (Our) achieves the highest pixel accuracy and DICE score.

Method	Pixel accuracy (↑)	Dice (↑)	HD (↓)
Original	0.898 ± 0.011	0.899 ± 0.009	84.251 ± 11.247
Reinhard [7]	0.897 ± 0.012	0.895 ± 0.013	91.470 ± 17.201
Macenko [8]	0.898 ± 0.006	0.899 ± 0.010	88.455 ± 12.533
Vahadane [9]	0.895 ± 0.013	0.895 ± 0.014	85.934 ± 9.536
StainGAN [15]	0.874 ± 0.009	0.873 ± 0.009	101.010 ± 14.506
SIST [20]	0.886 ± 0.009	0.884 ± 0.015	98.485 ± 16.436
SAGAN [44]	0.900 ± 0.012	0.896 ± 0.019	94.646 ± 16.765
CAGAN [22]	0.875 ± 0.011	0.873 ± 0.012	98.928 ± 15.151
MultiPathGAN [16]	0.898 ± 0.007	0.897 ± 0.013	86.849 ± 14.099
JeongSDE [23]	0.890 ± 0.008	0.889 ± 0.013	93.183 ± 12.150
Pix2Pix [45]	0.896 ± 0.006	0.896 ± 0.011	**81.778 ± 17.165**
Palette [31]	0.898 ± 0.010	0.895 ± 0.015	89.017 ± 17.212
SAStainDiff(Ours)	**0.903 ± 0.005**	**0.900 ± 0.016**	83.196 ± 14.817

abnormalities if the template image selection is not optimal. For example, the results of Reinhard tend to be overly bright and translucent. StainGAN and MultiPathGAN learn the cellular texture of the target domain, introduce incorrect modifications and alter the texture structure of cells. Moreover, SAGAN exhibits grid artifacts and Pix2Pix exhibits slight mosaic distortion. These factors collectively contribute to a decline in classifier performance, resulting in inferior classifier performance compared to traditional methods. Our proposed method effectively preserves the original tissue cell contrast while normalizing the stain to another color domain.

4.3. Stain normalization in the segmentation task

In the MICCAI'15 GlaS [36] segmentation task, we also standardize all images to the Aperio domain of MITOS-ATYPIA'14. Table 5 shows that stain normalization does not always improve the segmentation performance. Specifically, only the proposed method can improve in all three metrics: pixel accuracy, Dice score, and Hausdorff distance. One of the reasons for the limited effectiveness of stain normalization in improving the models is the limited stain styles in the MICCAI'15 GlaS dataset.

Another reason is that stain normalization results from some methods can compromise the integrity of the original images, leading to the loss of stain information or a degradation in image quality. For the enlarged windows in Fig. 7, Reinhard, StainGAN, SIST, CAGAN, and JeongSDE stain red blood cells to normal colors similar to the surrounding tissue, which is incorrect, resulting in the loss of some important structural information from the original RGB colors and weakening the clarity of cell boundaries. Severe artifacts are observed in the right half of the images with StainGAN and SAGAN. This indicates that although these methods can achieve stain normalization, they exhibit poorer generalization and are prone to generating erroneous information compared to our approach. Our proposed method effectively addresses these issues and performs well on this dataset.

It should be noted that Palette and our method employ the same set of model weights. After applying our rescheduled sampling strategy, it can improve inference speed while achieving similar segmentation results.

In Fig. 8, we present the Dice score with a threshold of 0.5. Among methods exhibiting significant segmentation errors such as Reinhard, StainGAN, and SAGAN, their Dice scores are lower compared to other methods. Our method shows segmentation results that closely resemble the Ground Truth, exhibit higher confidence, well-defined boundaries, and the highest dice score. This demonstrates the high effectiveness of our proposed stain normalization method, which achieves superior performance without the need for retraining on this dataset.

4.4. Ablation study

To verify the effectiveness of each design in the proposed method, we investigate the contributions of stain augmentation, Gaussian blur

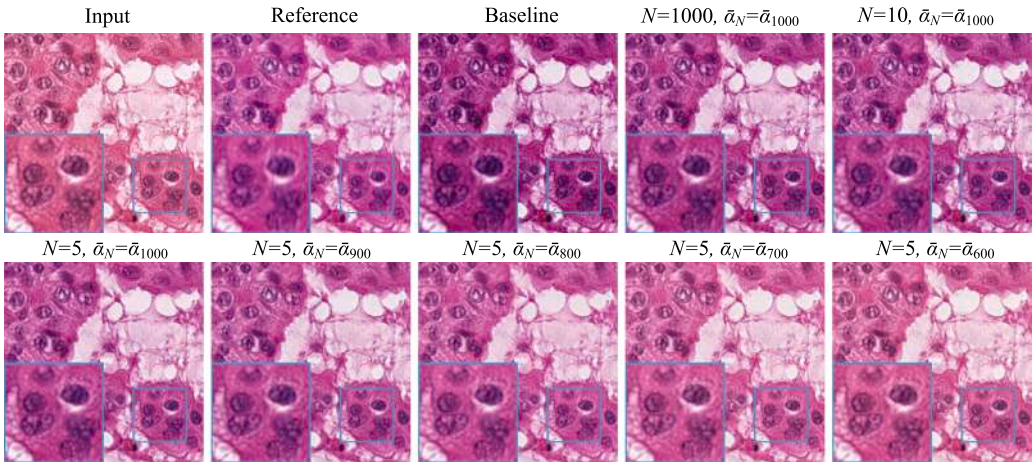


Fig. 9. Visual comparison of rescheduled sampling results, from the Aperio domain to the Hamamatsu domain. The image sample from pure noise exhibits a deep coloration. The staining becomes lighter with later initial points. $N = 5, \bar{\alpha}_{900}$ exhibits higher similarity with the reference image.

Table 6
Comparison of normalizing slices from Hamamatsu to Aperio performance on ablation study. "Pairwise" indicates training using paired data without any augmentation. "Aug W/o blur" signifies the use of only stain augmentation without adding Gaussian blur. "Aug" refers to stain augmentation with adding Gaussian blur.

Method	SSIM (↑)	PSNR (↑)	LPIPS (↓)
Pairwise	**0.828 ± 0.090**	**24.399 ± 1.880**	**0.065 ± 0.030**
Aug W/o blur	0.760 ± 0.094	20.631 ± 2.514	0.150 ± 0.044
Aug	0.761 ± 0.099	21.673 ± 1.644	0.132 ± 0.046

Table 7
Comparison of normalizing slices from Aperio to Hamamatsu performance on rescheduled sampling strategy study. "N" refers to the step size in sampling process and "$\bar{\alpha}_N$" refers to the initial sampling point.

Method	SSIM (↑)	PSNR (↑)	LPIPS (↓)
Baseline	0.781 ± 0.094	21.909 ± 2.148	0.131 ± 0.045
$N = 1000, \bar{\alpha}_N = \bar{\alpha}_{1000}$	0.783 ± 0.094	22.296 ± 2.133	0.130 ± 0.045
$N = 10, \bar{\alpha}_N = \bar{\alpha}_{1000}$	0.784 ± 0.093	23.067 ± 1.860	0.127 ± 0.043
$N = 5, \bar{\alpha}_N = \bar{\alpha}_{1000}$	0.790 ± 0.092	23.475 ± 1.812	0.124 ± 0.043
$N = 5, \bar{\alpha}_N = \bar{\alpha}_{900}$	**0.791 ± 0.092**	**23.523 ± 1.808**	**0.123 ± 0.043**
$N = 5, \bar{\alpha}_N = \bar{\alpha}_{800}$	0.791 ± 0.093	23.300 ± 1.690	0.124 ± 0.043
$N = 5, \bar{\alpha}_N = \bar{\alpha}_{700}$	0.790 ± 0.093	22.987 ± 1.576	0.126 ± 0.042
$N = 5, \bar{\alpha}_N = \bar{\alpha}_{600}$	0.789 ± 0.093	22.670 ± 1.472	0.129 ± 0.042

augmentation, and rescheduled sampling strategy for self-supervised learning on the MITOS-ATYPIA'14 dataset. The sampling strategy used in Table 6 is the original sampling strategy ($N = 1000$), which is consistent with Palette. Below are the experimental results and analysis of several modules we designed.

Stain augmentation. As shown in Table 6, paired training resulted in the highest metrics. However, in practical applications, acquiring fully paired data is quite challenging. In contrast, self-supervised learning exhibits greater generalization capability and can be applied to different datasets without retraining.

Gaussian blur augmentation. From Table 6, when employing Gaussian blur in augmentation, there is an increase in PSNR and LPIPS compared to the situation without Gaussian blur, demonstrating the effectiveness of Gaussian blur.

Rescheduled sampling strategy. The comparison of sampling from pure noise, as well as the variants of rescheduled sampling with different step sizes and initial sampling points, are shown in Table 7 and Fig. 9. With a step size of $N = 1000$ and a starting point from pure Gaussian noise (Baseline), the staining appears darker. When starting from a starting point of $\bar{\alpha}_N = \bar{\alpha}_{1000}$, the results seem to improve in quality. Different sampling step sizes yield similar results, such as $N = 10$ and $N = 5$. This further validates the effectiveness of our rescheduled sampling strategy.

Furthermore, by selecting different initial sampling points such as $\bar{\alpha}_{1000}$, $\bar{\alpha}_{900}$, $\bar{\alpha}_{800}$, $\bar{\alpha}_{700}$, and $\bar{\alpha}_{600}$, we observe that the choice of initial sampling points affects the intensity of staining. This demonstrates the flexibility of this strategy in controlling stain normalization results to adapt to various scenarios.

Considering the comprehensive evaluation, after applying stain augmentation, Gaussian blur augmentation, and rescheduled sampling strategy, our model achieves the best performance without paired training data.

4.5. Discussion

Fig. 10 visualizes the distribution of the color style of different datasets, using t-Distributed Stochastic Neighbor Embedding (t-SNE) [46]. From Fig. 10(a), samples from different scanners and tissue slices exhibit distinct clusters, indicating the diversity of datasets used in the experiment. From Fig. 10(b), the samples after our proposed method show an overlapping pattern, effectively demonstrating the robustness of our proposed method in handling various stain styles. This ability ensures normalizing different histopathological images to a target distribution in clinical practice.

To further demonstrate the adaptability of our proposed method to different stain styles without the need for retraining, Fig. 11 presents samples from various stain styles and cancer types in The Cancer Genome Atlas (TCGA). The visual results substantiate this claim, showcasing robustness even in the presence of artifacts.

In this paper, we propose SAStainDiff, a self-supervised stain normalization network based on the denoising diffusion probabilistic model with a stain augmentation training strategy and rescheduled sampling strategy. Firstly, we systematically reformulate the stain normalization task into a self-supervised formulation of pixel-wise color-to-color mapping task. This strategy has shown promising performance on image-to-image translation models, not only GANs but also DDPMs. Secondly, we simulate various stain styles and Gaussian blur of pathological images through stain augmentation. This approach is versatile and practical and does not require retraining when applied to different datasets. It eliminates the need for paired data, mitigating the high cost associated with data acquisition. Thirdly, we apply a rescheduled sampling strategy to the diffusion model to solve the stain normalization problem. This improves the sampling speed while maintaining the quality of stain normalization, compared with the original diffusion model.

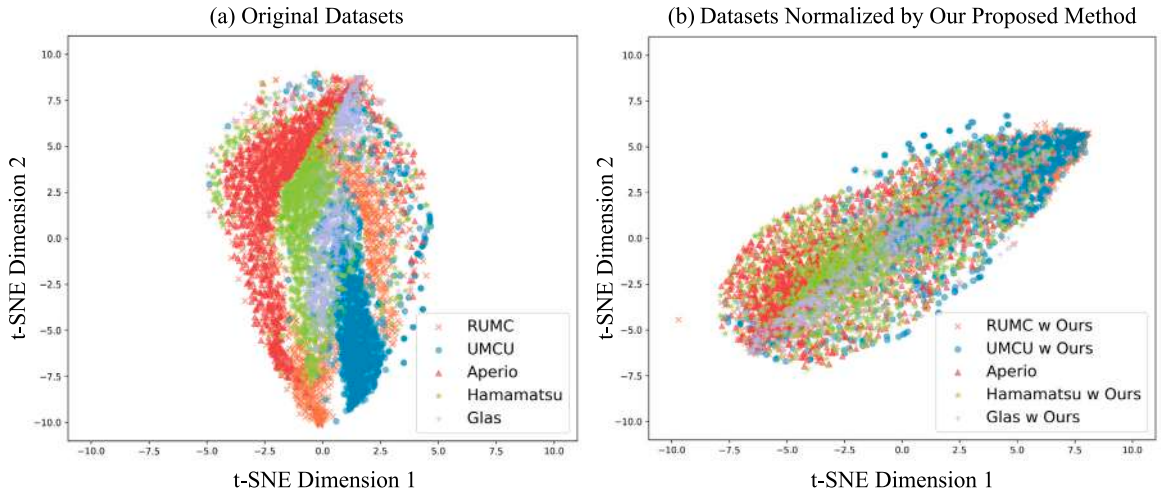


Fig. 10. The t-SNE visualization of five domains used in the experiment. (a) Samples from different domains exhibit clear distinctions. (b) We normalize images from four other domains to Aperio, where samples from different domains show highly similar t-SNE distributions after stain normalization. Validating the diversity of the data and the impact of our proposed method on histopathological images.

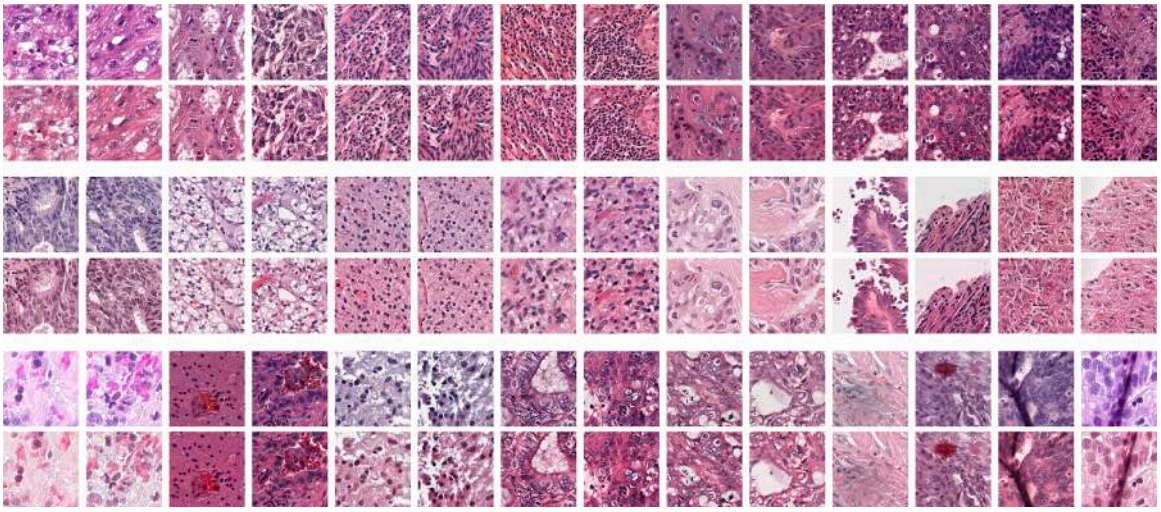


Fig. 11. Samples from different stain styles and disease types. In each group, the first row displays the original images, while the second row showcases the images normalized by our method. Some special cases that may be encountered in real clinical applications are shown in the third group. These visualization results highlight the capability of our method to perform stain normalization across diverse clinical data.

4.6. Limitation, prospect, and future work

Computational resources. In terms of computational resources, DDPMs suffer from the issue of long training and inference time, while GANs still hold an advantage. Although we have effectively improved the inference speed to approximately half that of GANs, with better robustness, substantial training resources are still required. However, there have been numerous methods proposed to accelerate DDPMs [47, 48], and we believe that DDPMs hold great potential for various applications.

Comprehensiveness. The versatility and adaptability of SAStainDiff across different datasets and stain styles without the need for retraining and paired data render it a valuable tool for analyzing digital pathology slice images with diverse tissue types and stain styles. Currently, we have only conducted quantitative analysis, classification, and segmentation experiments on a single publicly available dataset for each application. The comprehensiveness is limited, lacking an analysis of extreme stain scenarios.

Future directions. In the future, we plan to extend our approach to a broader range of cancer types and stain protocols to validate the generalization ability on other histopathological datasets and downstream tasks. Specifically, we plan to integrate SAStainDiff into existing pathology systems, which can simplify clinical workflows by reducing the time and resources required for manual color adjustments, thus enhancing efficiency in pathology laboratories and diagnostic settings.

5. Conclusion

In summary, this paper proposes SAStainDiff for stain normalization of histological images. It circumvents the drawbacks of colorization-based methods that result in the loss of color information. With a rescheduled sampling strategy, it improves the sampling speed of the model, enables a certain degree of controllability over the staining intensity, and enhances the model's generalization ability. Compared with existing methods, SAStainDiff can perform well on images with different stain styles without retraining. We believe this algorithm can enhance CAD diagnostic accuracy and aid pathologists in making more accurate and confident diagnostic decisions by providing a more standardized and controllable approach to stain normalization.

CRediT authorship contribution statement

Huaishui Yang: Writing – review & editing, Writing – original draft, Software, Methodology. **Mengye Lyu:** Writing – review & editing, Supervision. **Shiyue Yan:** Software, Methodology. **Tianzhao Zhong:** Resources, Data curation. **Jihao Li:** Validation, Investigation, Formal analysis. **Tong Xu:** Validation, Formal analysis. **Huhan Xie:** Resources, Data curation. **Shaojun Liu:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.bspc.2025.107861.

Data availability

The data, code, and pre-trained model weights are publicly available at: https://github.com/yhuaishui/SAStainDiff.

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